

VEX COMPETITIVE RATING

A strength-based ranking system for VEX V5 teams and events

VCR combines opponent-adjusted match results, scoring contribution, defense, autonomous performance and expectation-adjusted event finishes. Awards are excluded.

TEAM RATINGS / EVENT TIERS / EVENT WEIGHT

PURPOSE

1. System overview

The VEX Competitive Rating (VCR) estimates a team's current competitive strength. It is not a participation score and does not use judged awards. Every update asks one question: **did this team perform better or worse than a team of its rating was expected to perform?**

A team begins near **1500**. Its rating changes after each match and event. Strong results against strong opposition matter more than routine wins against weak opposition. Event Weight adds a second layer: elite events offer greater upside, while poor performances by favorites at weak events carry greater downside.

Design principles

Principle	Rule
Strength, not attendance	Extra events cannot create rating points without above-expected performance.
Opponent-aware	Results are evaluated against alliance and opponent strength.
Game-aware	Statistics are normalized within the current game and event environment.
No awards	Judged awards and skills unrelated to match competition are excluded.
Transparent	Every rating change can be decomposed into matches, contribution, autonomous and finish.
Predictive	Weights should be tuned by forecasting later matches, not by matching reputation.

Rating architecture

Layer	Output	Timing
Match model	Opponent-adjusted rating change	After each official match
Performance model	OPR/DPR/CCWM and autonomous residuals	After event data is complete
Event model	Tier, Event Weight and finish surprise	Tier frozen before event; result after event
Uncertainty model	Update size and confidence	Continuously

TEAM MODEL

2. Competitive inputs

Component	Target share	Measures
Match performance	40%	Wins, ties, losses, margin and opponent strength
Adjusted OPR	20%	Estimated scoring contribution
Adjusted DPR	10%	Estimated points allowed or defensive impact
CCWM	10%	Estimated net contribution to score margin
Autonomous	10%	AWP rate and estimated autonomous contribution
Event performance	10%	Qualification and elimination finish versus expectation

The shares are starting parameters, not permanent truths. Backtesting should optimize them for prediction accuracy while maintaining interpretability.

Avoiding double-counting

OPR, DPR and CCWM are mathematically related. In many implementations, **CCWM is approximately OPR minus DPR**. Adding all three at full strength would count the same signal more than once. Build a residualized contribution score instead:

$$\text{Contribution} = 0.55 \text{ Z(OPR)} + 0.20 \text{ Z(-DPR)} + 0.25 \text{ Z(CCWM)}$$

$$\text{Z}(x) = (x - \text{event mean}) / \text{event standard deviation}$$

If the data provider defines higher DPR as better defense, remove the negative sign. The project must document and preserve one convention.

Better long-term estimator

Estimate OPR-family values with ridge regression rather than unregularized least squares. Ridge regression reduces extreme estimates when teams play few matches or repeatedly share partners. Store standard errors so low-confidence statistics receive less influence.

Season and recency

Never compare raw scoring statistics across different games. Use current-season normalized values. Apply gradual recency decay to old evidence - for example, a 90-day half-life - while retaining rating uncertainty rather than resetting established teams abruptly.

OPPONENT ADJUSTMENT

3. Match rating

For alliance AB against alliance CD, begin with the mean pre-match rating of each alliance:

$$R(\text{red}) = (R(A) + R(B))/2 \quad R(\text{blue}) = (R(C) + R(D))/2$$

$$E(\text{red}) = 1 / (1 + 10^{((R(\text{blue}) - R(\text{red}))/400)})$$

Actual score is 1.0 for a win, 0.5 for a tie and 0.0 for a loss. The unscaled surprise is Actual minus Expected.

Margin of victory

$$MOV = 1 + 0.35 \tanh(\text{ScoreMargin} / \text{EventScoreSD})$$

$$\text{DeltaMatch} = K_{\text{match}} \times (\text{Actual} - \text{Expected}) \times MOV$$

Start with **Kmatch = 20**. The bounded tanh function lets dominant performances matter without allowing a single blowout to overwhelm the model. Exclude disqualifications or score corrections only according to a documented data policy.

Sharing alliance credit

Allocate alliance change using predicted individual contribution, clamped so neither partner receives less than 35% or more than 65%:

$$\text{Share}(i) = \text{clamp}(0.35, \text{PredContribution}(i) / \text{SumAllianceContribution}, 0.65)$$

When individual estimates are missing or unreliable, split the update 50/50. This avoids inventing precision.

Uncertainty

Maintain a rating deviation for every team. New, recently rebuilt, or inactive teams have greater uncertainty and therefore larger updates. Established teams have smaller, more stable changes. Publish both rating and confidence, but rank primarily by the conservative estimate:

$$\text{DisplayedStrength} = \text{Rating} - 1.0 \times \text{RatingDeviation}$$

PERFORMANCE ADJUSTMENT

4. Autonomous and event finish

Autonomous performance should contain both outcome and contribution. Because AWP is earned by an alliance, compare the actual AWP outcome with its predicted probability rather than treating every AWP as equal.

$$\text{AutoScore} = 0.60 \text{ Z(Adjusted AWP Rate)} + 0.40 \text{ Z(Auto Contribution)}$$

Adjusted AWP Rate is the residual between observed AWP outcomes and probabilities predicted from all four participating teams. Auto Contribution may be estimated only when scoring breakdowns are reliable; otherwise use the AWP residual alone and lower confidence.

Event result percentile

$$\text{ResultPercentile} = 0.55 \text{ QualificationPercentile} + 0.45 \text{ EliminationPercentile}$$

Elimination result	Suggested percentile
Champion	1.00
Finalist	0.90
Semifinalist	0.75
Quarterfinalist	0.55
Did not qualify	Use qualification percentile

Expected percentile is calculated from the team's frozen pre-event rating relative to the field. The model then measures surprise:

$$\text{EventSurprise} = \text{ResultPercentile} - \text{ExpectedPercentile}$$

A favorite that finishes where expected receives little change. An underdog that reaches the final of an elite field receives a large positive surprise. Qualification percentile should use official ranking positions; elimination percentile should account for bracket size and byes.

Alliance selection caution

Elimination advancement is partly determined by alliance selection. Do not attribute an entire alliance finish equally. Combine finish surprise with match-level contribution and cap the finish component at 10% of the total event update.

FIELD QUALITY

5. Event ranking

An event's tier is determined from the registered field's **frozen pre-event ratings**. Branding alone cannot make an event elite. Freezing the inputs prevents circular logic in which strong performances retroactively make their own event stronger.

Event Strength Score (ESS)

$$\text{ESS} = 0.35 \text{ FieldStrength} + 0.30 \text{ FieldDepth} + 0.15 \text{ EliteDensity} + 0.10 \text{ RegionalDiversity} + 0.10 \text{ FormatQuality}$$

Factor	Weight	Definition
Field Strength	35%	Percentile rank of the field's mean pre-event rating
Field Depth	30%	60% top-quarter strength + 40% top-half strength
Elite Density	15%	Normalized density of global top 1%, 5% and 10% teams
Regional Diversity	10%	Strong-team representation across regions/countries
Format Quality	10%	Matches per team, field size, complete data and valid bracket

Field Depth prevents one famous team from carrying an otherwise weak event. Regional Diversity distinguishes genuinely international fields without automatically penalizing a deep regional championship. Format Quality measures competitive reliability only; it does not include judged awards.

Eligibility rules

Rule	Recommended policy
Minimum size	24 teams for an official tier; smaller events remain provisional
Data completeness	At least 95% of official match results present
Tier freeze	Freeze 24 hours before the first scheduled match
Late roster changes	Recalculate only if more than 10% of teams change
Regional ceiling	Ordinary regional events may not exceed Bronze S
Signature range	Signature Events may be Bronze S, Silver S or Gold S

TIER SYSTEM

6. Event tiers and Event Weight

Tier	ESS	Event Weight	Typical classification
C	0-34	0.70	Small or weak local competition
B	35-49	0.82	Strong local / average regional competition
A	50-64	0.95	Strong regional competition
Bronze S	65-74	1.15	Regional championship / entry Signature
Silver S	75-87	1.35	Strong Signature Event
Gold S	88-100	1.60	Elite international Signature Event
Worlds	Special	1.85	VEX World Championship

If the public taxonomy must end at Gold S, display Worlds as Gold S with a championship modifier of 1.85.

Asymmetric Expectation Weight

A simple symmetric multiplier fails the design goal: lowering the weight of a C event would also reduce the penalty for a favorite that fails there. Use an asymmetric multiplier on event surprise:

$$\text{ContextWeight} = \text{EventWeight}, \text{ if Surprise} > 0; 1/\text{EventWeight}, \text{ if Surprise} < 0$$

$$\text{DeltaEvent} = \text{Kevent} \times \text{EventSurprise} \times \text{ContextWeight} \times \text{Reliability}$$

Start with **Kevent = 80** and cap the total event-level adjustment at **+/-80 rating points**. Reliability ranges from 0 to 1 and reflects match count, data completeness and statistical uncertainty.

Worked comparison

Result	C (0.70)	Gold S (1.60)	Interpretation
+0.20 surprise	+0.140 scaled	+0.320 scaled	Elite-event overperformance earns more
-0.20 surprise	-0.286 scaled	-0.125 scaled	Weak-event failure by a favorite hurts more

IMPLEMENTATION

7. Final update and safeguards

At event completion, combine the independently calculated components:

$$\text{DeltaVCR} = 0.65 \text{ DeltaMatch} + 0.20 \text{ DeltaContribution} + 0.10 \text{ DeltaAuto} + 0.05 \text{ DeltaFinish}$$

The 65/20/10/5 blend is preferred over the conceptual 40/20/10/10/10 input shares for implementation because match results already absorb some scoring, defense and finish information. This more conservative blend reduces double-counting. Tune it through backtesting.

Required safeguards

Control	Purpose
Chronological processing	Never allow future results to grade an earlier event.
Frozen pre-event ratings	Prevent circular event strength and expectation calculations.
Season normalization	Never compare raw OPR-family values across different games.
Positive low-tier limit	Only the best two positive C/B event bonuses count in a rolling 90-day window.
All negative results count	Teams cannot hide weak performances through selective participation.
Event cap	Limit event-level movement to +/-80 points.
Data-quality factor	Reduce influence when schedules, scores or autonomous data are incomplete.
Public decomposition	Show old rating, each component, Event Weight and new rating.

Validation plan

Backtest chronologically across at least one complete season. Optimize for log loss, Brier score and calibration of predicted match probabilities. Compare VCR against win rate, plain Elo and OPR-only baselines. Validate separately by region, event tier and team experience. Publish major parameter changes and never tune on the same season used for final evaluation.

Recommended public display

Field	Example
World rank	#12
VCR	1718
Confidence	+/-42
Recent change	+31
Event tier	Silver S (1.35)
Reason	+18 matches, +7 contribution, +4 auto, +2 finish

Status: initial specification. The constants in this document are defensible starting values; they should become data-calibrated parameters after historical VEX match data is available.